

8. $\frac{-1 \pm \sqrt{5}}{2}, \frac{-1 \pm \sqrt{5}}{2}$

Hint. $x(x+2)(x^2-1) = -1 \Rightarrow x(x+2)(x+1)(x-1) = -1$

$$\Rightarrow [x(x+1)][(x+2)(x-1)] = -1$$

$$\Rightarrow (x^2+x)(x^2+x-2) = -1$$

$$\Rightarrow y(y-2) = -1 \text{ where } y = x^2+x$$

$$\Rightarrow y^2 - 2y + 1 = 0 \Rightarrow y = 1, 1$$

$$\Rightarrow x^2+x = 1, x^2+x = 1.$$

9. $2, -6, \frac{-4 \pm i\sqrt{59}}{2}$ **Hint.** L.H.S. = $4[(x+1)(x+3)] \left[\left(x + \frac{3}{2}\right) \left(x + \frac{5}{2}\right) \right]$.

10. $0, 1, \frac{1 \pm \sqrt{57}}{2}$ **Hint.** L.H.S. = $(x+3)(x-4)(x-2)(x+1)$.

11. (i) $\frac{9}{13}, \frac{4}{13}$ (ii) $-\frac{5}{16}$

12. (i) 4 (ii) 1, 5 13. (i) No solution (ii) $\frac{37}{4}$

14. (i) $3, -\frac{5}{3}$ (ii) 5, 6

Hint. (ii) Given equation is $\sqrt{3x^2 - 7x - 30} = (x-5) + \sqrt{2x^2 - 7x - 5}$.

Square both sides,

$$3x^2 - 7x - 30 = (x-5)^2 + (2x^2 - 7x - 5) + 2(x-5)\sqrt{2x^2 - 7x - 5}$$

$$\Rightarrow 10x - 50 - 2(x-5)\sqrt{2x^2 - 7x - 5} = 0$$

$$\Rightarrow 2(x-5)(5 - \sqrt{2x^2 - 7x - 5}) = 0.$$

15. $6, -\frac{17}{3}$ 16. $\pm \sqrt{\frac{5}{3}}$

17. (i) $\frac{5 \pm \sqrt{29}}{2}, \frac{-3 \pm \sqrt{13}}{2}$ (ii) $\frac{1}{2}, \frac{2}{3}, \frac{3}{2}, 2$

18. $1 \pm i, 2 \pm \sqrt{2}$ **Hint.** Divide throughout by x^2 .

5.3 NATURE OF ROOTS OF A QUADRATIC EQUATION

The roots of the equation $ax^2 + bx + c = 0, a \neq 0$, are

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \text{ where } a, b, c \in \mathbf{R}.$$

Thus the nature of the roots depends on the quantity under the square root sign i.e. $b^2 - 4ac$. This quantity is called *discriminant* of the equation and is usually denoted by Δ .

$$\therefore \text{Discriminant } \Delta = b^2 - 4ac.$$

Case I. When a, b, c are real numbers, $a \neq 0$.

(i) If $\Delta = b^2 - 4ac = 0$, then roots are **equal** (and real).

(ii) If $\Delta = b^2 - 4ac > 0$, then roots are **real and unequal**.

(iii) If $\Delta = b^2 - 4ac < 0$, then roots are **complex**. It is easy to see that roots are a pair of **complex conjugates**.

Case II. When a, b, c are rational numbers, $a \neq 0$.

(i) If $\Delta = b^2 - 4ac = 0$, then roots are **rational and equal**.

(ii) If $\Delta = b^2 - 4ac > 0$, and Δ is a perfect square of a rational number, then roots are **rational and unequal**.

(iii) If $\Delta = b^2 - 4ac > 0$ but Δ is not a square of rational number, then roots are **irrational and unequal**. These form a pair of irrational conjugates $p + \sqrt{q}, p - \sqrt{q}$ where $p, q \in \mathbb{Q}, q > 0$.

(iv) If $\Delta = b^2 - 4ac < 0$, then roots are a pair of complex conjugates.

ILLUSTRATIVE EXAMPLES

Example 1. Discuss the nature of the roots of the following equations :

(i) $4x^2 - 12x + 9 = 0$

(ii) $3x^2 - 10x + 3 = 0$

(iii) $9x^2 - 2 = 0$

(iv) $x^2 + x + 1 = 0$.

Solution. (i) Here the coefficients are rational, and

$$\text{discriminant } \Delta = b^2 - 4ac = (-12)^2 - 4(4)(9) = 144 - 144 = 0.$$

Hence, the roots are rational and equal.

(ii) Here the coefficients are rational, and

$$\text{discriminant } \Delta = b^2 - 4ac = (-10)^2 - 4(3)(3) = 100 - 36 = 64.$$

Now $\Delta = 64 > 0$, and 64 is a perfect square of a rational number.

Hence, the roots are rational and unequal.

(iii) Here the coefficients are rational, and

$$\text{discriminant } \Delta = b^2 - 4ac = (0)^2 - 4(9)(-2) = 72.$$

Now $\Delta = 72 > 0$ but is not a perfect square of a rational number.

Hence, the roots are irrational and unequal. These form a pair of irrational conjugates.

(iv) Here the coefficients are rational, and

$$\text{discriminant } \Delta = b^2 - 4ac = (1)^2 - 4(1)(1) = -3 < 0.$$

Hence, the roots are a pair of complex conjugates.

Example 2. Discuss the nature of the roots of the following equations:

(i) $4x^2 - 4\sqrt{3}x + 3 = 0$

(ii) $2\sqrt{3}x^2 - 5x + \sqrt{3} = 0$.

Solution. (i) Here the coefficients are real, and

$$\text{discriminant } \Delta = b^2 - 4ac = (4\sqrt{3})^2 - 4(4)(3) = 48 - 48 = 0.$$

Hence, the roots are real and equal.

(ii) Here the coefficients are real, and

$$\text{discriminant } \Delta = b^2 - 4ac = (-5)^2 - 4(2\sqrt{3})(\sqrt{3}) = 25 - 24 = 1 > 0.$$

Hence, the roots are real and unequal.

Example 3. If $a, b, h \in \mathbb{R}$, show that the equation $(x-a)(x-b) = h^2$ has real roots:

Solution. The given equation can be written as

$x^2 - (a+b)x + (ab-h^2) = 0$, which is a quadratic equation with real coefficients.

$$\text{Its discriminant} = (-(a+b))^2 - 4 \cdot 1 \cdot (ab-h^2)$$

$$= (a+b)^2 - 4ab + 4h^2$$

$$= (a-b)^2 + (2h)^2 \geq 0$$

$\Rightarrow$ the given equation has real roots.

Example 4. If the roots of $ax^2 + x + b = 0$ are real and distinct, show that the roots of the equation

$$\frac{x^2 + 1}{x} = 4\sqrt{ab} \text{ are imaginary.}$$

Solution. Given roots of the equation $ax^2 + x + b = 0$ are real and distinct

$$\Rightarrow \text{discriminant} > 0 \Rightarrow (1)^2 - 4ab > 0$$

$$\Rightarrow 4ab - 1 < 0 \quad \dots(i)$$

$$\text{Now } \frac{x^2 + 1}{x} = 4\sqrt{ab} \Rightarrow x^2 + 1 = 4\sqrt{ab} x$$

$$\Rightarrow x^2 - 4\sqrt{ab} x + 1 = 0 \quad \dots(ii)$$

$$\text{Its discriminant} = (-4\sqrt{ab})^2 - 4 \times 1 \times 1 = 16ab - 4$$

$$= 4(4ab - 1) < 0$$

(using (i))

$\Rightarrow$ the roots of equation (ii) are imaginary (complex).

Hence, the roots of the equation $\frac{x^2 + 1}{x} = 4\sqrt{ab}$ are imaginary.

Example 5. Given that $a, b, c \in \mathbb{Q}$, show that the roots of the equation

$$(b - c)x^2 + (c - a)x + (a - b) = 0$$

are rational. Find the condition for the roots to be equal.

Solution. Here coefficients are rational and

$$\begin{aligned} \text{discriminant } \Delta &= (c - a)^2 - 4(b - c)(a - b) \\ &= c^2 + a^2 - 2ac - 4(ab - b^2 - ac + bc) \\ &= (c^2 + a^2 + 2ac) + 4b^2 - 4b(a + c) \\ &= (c + a)^2 - 4b(a + c) + (2b)^2 = (c + a - 2b)^2. \end{aligned}$$

Thus Δ is the square of a rational number.

Hence, the roots of the given equation are rational.

For roots to be equal, $\Delta = (c + a - 2b)^2 = 0$

$$\Rightarrow c + a - 2b = 0 \Rightarrow b = \frac{a + c}{2}, \text{ which is the required condition.}$$

Example 6. Discuss the nature of the roots of the equation

$$(m + 6)x^2 + (m + 6)x + 2 = 0.$$

Solution. Discriminant $\Delta = (m + 6)^2 - 4(m + 6)(2)$

$$= m^2 + 12m + 36 - 8m - 48$$

$$= m^2 + 4m - 12 = (m + 6)(m - 2).$$

Now (i) Roots are real and equal if $\Delta = (m + 6)(m - 2) = 0$

i.e. if $m = 2$ or -6 .

(ii) Roots are real and unequal when $\Delta = (m + 6)(m - 2) > 0$

i.e. when $m < -6$ or when $m > 2$.

(iii) Roots are a pair of complex conjugates when

$$\Delta = (m + 6)(m - 2) < 0$$

i.e. when $-6 < m < 2$.

Example 7. For what value of k does the expression $x^2 - 2x(1 + 3k) + 7(3 + 2k)$ become a perfect square? Also write the expression as perfect square in that case.

Solution. The given expression will be a perfect square iff the equation

$$x^2 - 2x(1 + 3k) + 7(3 + 2k) = 0 \text{ has equal roots}$$

i.e. iff the discriminant of this equation = 0 i.e. iff $\Delta = 0$

$$\Leftrightarrow (-2(1+3k))^2 - 4(1)(7(3+2k)) = 0$$

$$\Leftrightarrow 9k^2 - 8k - 20 = 0$$

$$\Leftrightarrow k = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(9)(-20)}}{2 \times 9} = \frac{8 \pm 28}{18} = 2, -\frac{10}{9}.$$

When $k = 2$, the given expression

$$= x^2 - 2x(1 + 3 \times 2) + 7(3 + 2 \times 2)$$

$$= x^2 - 14x + 49 = (x - 7)^2.$$

When $k = -\frac{10}{9}$, the given expression

$$= x^2 - 2x \left[1 + 3 \left(-\frac{10}{9} \right) \right] + 7 \left[3 + 2 \left(-\frac{10}{9} \right) \right]$$

$$= x^2 + \frac{14}{3}x + \frac{49}{9} = \left(x + \frac{7}{3} \right)^2.$$

Example 8. Determine a positive real value of k such that both the equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ may have real roots.

Solution. The given equations are

$$x^2 + kx + 64 = 0 \text{ and } x^2 - 8x + k = 0.$$

As both the equations have real roots, the discriminant of each ≥ 0

$$\Rightarrow k^2 - 4 \times 1 \times 64 \geq 0 \text{ and } (-8)^2 - 4 \times 1 \times k \geq 0$$

$$\Rightarrow k^2 - 256 \geq 0 \text{ and } 64 - 4k \geq 0$$

$$\Rightarrow (k + 16)(k - 16) \geq 0 \text{ and } 16 - k \geq 0$$

$$\Rightarrow (k \leq -16 \text{ or } k \geq 16) \text{ and } k \leq 16 \text{ but } k \text{ is positive real number}$$

$$\Rightarrow k \geq 16 \text{ and } k \leq 16 \Rightarrow k = 16.$$

Hence, the positive real value of k for which both the given equations have real roots is 16.

Example 9. Show that if p, q, r, s are real numbers and $pr = 2(q + s)$ then atleast one of the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ has real roots.

Solution. If possible, let both the equations have complex roots. Then, for both, discriminant < 0

$$\Rightarrow p^2 - 4q < 0, r^2 - 4s < 0$$

$$\Rightarrow (p^2 - 4q) + (r^2 - 4s) < 0 \Rightarrow p^2 + r^2 - 4(q + s) < 0$$

$$\Rightarrow p^2 + r^2 - 2pr < 0$$

(putting $2(q + s) = pr$)

$$\Rightarrow (p - r)^2 < 0, \text{ which is impossible, as both } p, r \text{ are real numbers}$$

$$\Rightarrow \text{our assumption is wrong}$$

$$\Rightarrow \text{atleast one of the given equations has real roots.}$$

EXERCISE 5.3

Very short answer/objective questions (1 to 2):

1. Find the nature of roots of the following equations without solving them:

(i) $x^2 + 9 = 0$

(ii) $4x^2 - 24x + 35 = 0$

(iii) $x^2 - 2\sqrt{2}x + 1 = 0$

(iv) $2x^2 - 2\sqrt{5}x + 3 = 0.$

2. (i) Prove that the roots of the equation $x^2 - 2ax + a^2 - b^2 - c^2 = 0$ are always real, $a, b, c \in \mathbb{R}$.

- (ii) Show that roots of the equation $(x - a)(x - b) = abx^2$; $a, b \in \mathbf{R}$ are always real. When are they equal?

Short answer questions (3 to 6):

3. Show that the roots of the equation $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$, where $a, b, c \in \mathbf{R}$ are always real. Find the condition that the roots may be equal. What are the roots when this condition is satisfied?
4. Discuss the nature of roots of the following equations :

$$(i) \sqrt{3}x^2 - 2x - \sqrt{3} = 0 \quad (ii) x^2 - (p + 1)x + p = 0 \quad (iii) (x - a)(x - b) = ab.$$

It is given that $p \in \mathbf{Q}$, and $a, b \in \mathbf{R}$.

5. Find m so that roots of the equation $(4 + m)x^2 + (m + 1)x + 1 = 0$ may be equal.
6. Show that the roots of the equation $x^2 + 2(3a + 5)x + 2(9a^2 + 25) = 0$ are complex unless $a = \frac{5}{3}$.

Long answer questions (7 to 14):

7. If $a, b, c, d \in \mathbf{R}$, show that the roots of the equation $(a^2 + c^2)x^2 + 2(ab + cd)x + (b^2 + d^2) = 0$ cannot be real unless they are equal.
8. If $a, b, c \in \mathbf{Q}$ and $a + b + c = 0$, show that roots of the equation $ax^2 + bx + c = 0$ are rational. Is the converse of above true?
9. Prove that the equation $x^2 + px - 1 = 0$ has real and distinct roots for all real values of p .
10. Show that the roots of the equation $x^2 - 2\left(m + \frac{1}{m}\right)x + 3 = 0$ are real for all (non-zero) real values of m .
11. If the roots of the equation $ax^2 + 2cx + b = 0$ are real and distinct, then show that the roots of the equation $x^2 - 2(a + b)x + a^2 + b^2 + 2c^2 = 0$ are non-real complex numbers.
12. If the roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are equal, show that $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$.
13. For what value of k , $(4 - k)x^2 + 2(k + 2)x + (8k + 1)$, is a perfect square.
14. If the roots of the equation $x^2 - 8x + m(m - 6) = 0$ are real and distinct, then find all possible values of m .

ANSWERS

- (i) pair of complex conjugates (ii) rational and unequal

(iii) real and unequal (iv) pair of complex conjugates
- (ii) Roots are real and equal when $a = b = 0$ **Hint.** (ii) $\Delta = (a - b)^2 + (2ab)^2$.
- Roots are equal when $a = b = c$. Then roots are a, a

Hint. $\Delta = 2((a - b)^2 + (b - c)^2 + (c - a)^2)$.
- (i) real and distinct

(ii) rational and distinct when $p \neq 1$; when $p = 1$, roots are rational and equal

(iii) real and distinct when $a + b \neq 0$; when $a + b = 0$, roots are both 0

5. 5, -3

8. Converse is not true. For example, $x^2 - x - 2 = 0$ has rational roots 2, -1 but sum of its coefficients $\neq 0$

13. 0, 3

14. $-2 < m < 8$

5.4 RELATIONS BETWEEN ROOTS AND COEFFICIENTS

Sum and Product of Roots

Let the roots of the equation $ax^2 + bx + c = 0$, $a \neq 0$, be

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Then, sum of roots $= \alpha + \beta$

$$= \frac{-b + \sqrt{b^2 - 4ac}}{2a} + \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \frac{-2b}{2a} = -\frac{b}{a}.$$

$$\therefore \text{Sum of roots} = \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}.$$

$$\begin{aligned} \text{Product of roots} = \alpha \cdot \beta &= \frac{-b + \sqrt{b^2 - 4ac}}{2a} \times \frac{-b - \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{(-b)^2 - (\sqrt{b^2 - 4ac})^2}{(2a)^2} = \frac{b^2 - (b^2 - 4ac)}{4a^2} = \frac{4ac}{4a^2} = \frac{c}{a}. \end{aligned}$$

$$\therefore \text{Product of roots} = \alpha \beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}.$$

Symmetric Functions of Roots

If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, then the expressions of the form $\alpha + \beta$, $\alpha\beta$, $\alpha^2 \pm \beta^2$, $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ etc. are called **functions** of the roots α and β .

If an expression doesn't change on interchanging α and β , then it is called **symmetric**. Thus $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$

is a symmetric function while $\alpha^2 - \beta^2$ is not a symmetric function. The expressions $\alpha + \beta$ and $\alpha\beta$ are called **elementary symmetric functions**.

We have already seen that for the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, the value of $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$. With this information, values of other functions of α and β can be calculated :

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta,$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta,$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta),$$

$$\alpha^4 + \beta^4 = [(\alpha + \beta)^2 - 2\alpha\beta]^2 - 2(\alpha\beta)^2 \text{ and}$$

$$\alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta) \text{ etc.}$$

5.4.1 Formation of a Quadratic Equation with Given Roots

Let the given roots be α and β . Suppose that the equation whose roots are α, β is $ax^2 + bx + c = 0$, $a \neq 0$.

Then $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$.

Now $ax^2 + bx + c = 0 \Leftrightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$ ($\because a \neq 0$)

$\Leftrightarrow x^2 - \left(-\frac{b}{a}\right)x + \left(\frac{c}{a}\right) = 0 \Leftrightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$\Leftrightarrow x^2 - (\text{sum of roots})x + \text{product of roots} = 0$

$\Leftrightarrow x^2 - Sx + P = 0$, where $S = \text{sum of roots}$ and $P = \text{product of roots}$.

5.4.2 To find conditions when roots are connected by given Relations

Sometimes the relation between roots of a quadratic equation is given and we are asked to find the condition i.e. relation between the coefficients a, b, c of quadratic equation. This is easily done using the formula $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$. This will be clear when you go through illustrative examples.

One common root

Suppose the equations $a_1x^2 + b_1x + c_1 = 0$ ($a_1 \neq 0$) and

$a_2x^2 + b_2x + c_2 = 0$ ($a_2 \neq 0$) have a common root, say α .

Then $a_1\alpha^2 + b_1\alpha + c_1 = 0$ and

$a_2\alpha^2 + b_2\alpha + c_2 = 0$.

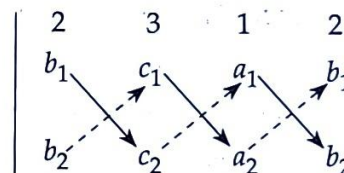
Solving by cross multiplication rule,

$$\frac{\alpha^2}{b_1c_2 - b_2c_1} = \frac{\alpha}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

From first two, $\alpha = \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$ and

from last two, $\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$.

Thus $\frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1} = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$



i.e. $(c_1a_2 - c_2a_1)^2 = (b_1c_2 - b_2c_1)(a_1b_2 - a_2b_1)$, which is the required condition for the two equations to have a common root.

Both roots common

Suppose the equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have both roots common. Let those roots be α, β .

Then, as α, β are roots of $a_1x^2 + b_1x + c_1 = 0$, $\alpha + \beta = -\frac{b_1}{a_1}$, $\alpha\beta = \frac{c_1}{a_1}$.

Similarly, $\alpha + \beta = -\frac{b_2}{a_2}$, $\alpha\beta = \frac{c_2}{a_2}$

$\Rightarrow -\frac{b_1}{a_1} = -\frac{b_2}{a_2}$, $\frac{c_1}{a_1} = \frac{c_2}{a_2} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, which is the required condition for the two

equations to have both roots common.

ILLUSTRATIVE EXAMPLES

Example 1. If α, β are roots of the equation $x^2 - 4x + 2 = 0$, find the values of

(i) $\alpha^2 + \beta^2$ (ii) $\alpha^2 - \beta^2$ (iii) $\alpha^3 + \beta^3$ (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution. As α, β are roots of the equation $x^2 - 4x + 2 = 0$,

$$\alpha + \beta = -\frac{(-4)}{1} = 4, \alpha\beta = \frac{2}{1} = 2.$$

$$(i) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (4)^2 - 2 \times 2 = 12.$$

$$(ii) \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta).$$

$$\text{Now } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = (4)^2 - 4 \times 2 = 8$$

$$\Rightarrow \alpha - \beta = \pm\sqrt{8} = \pm 2\sqrt{2}$$

$$\therefore \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta) = 4 \times (\pm 2\sqrt{2}) = \pm 8\sqrt{2}.$$

$$(iii) \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (4)^3 - 3 \times 2 \times 4 = 64 - 24 = 40.$$

$$(iv) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{4}{2} = 2.$$

Example 2. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then show that

$$ax^2 + bx + c = a(x - \alpha)(x - \beta).$$

Solution. Since α, β are roots of the equation $ax^2 + bx + c = 0$,

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}.$$

$$\begin{aligned} \text{Now } ax^2 + bx + c &= a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right] = a\left[x^2 - \left(-\frac{b}{a}\right)x + \frac{c}{a}\right] \\ &= a[x^2 - (\alpha + \beta)x + \alpha\beta] = a(x - \alpha)(x - \beta). \end{aligned}$$

Example 3. For the quadratic equation $(k - 1)x^2 = kx - 1, k \neq 1$, find k so that

(i) one root is -3

(ii) the sum of the roots is 2

(iii) the product of the roots is -3

(iv) the roots are equal

(v) the roots are numerically equal but opposite in sign.

Solution. The given quadratic equation is $(k - 1)x^2 - kx + 1 = 0$.

(i) Since -3 is a root of the given equation, it must satisfy the equation.

$$\therefore (k - 1)(-3)^2 - k(-3) + 1 = 0$$

$$\Rightarrow 9k - 9 + 3k + 1 = 0 \Rightarrow 12k = 8$$

$$\Rightarrow k = \frac{8}{12} = \frac{2}{3}.$$

(ii) Given the sum of the roots $= 2$

$$\Rightarrow -\frac{-k}{k - 1} = 2 \Rightarrow 2k - 2 = k \Rightarrow k = 2.$$

(iii) Given the product of the roots $= -3$

$$\Rightarrow \frac{1}{k - 1} = -3 \Rightarrow 1 = -3k + 3 \Rightarrow 3k = 2 \Rightarrow k = \frac{2}{3}.$$

(iv) Given the roots are equal $\Rightarrow$ discriminant $\Delta = 0$

$$\Rightarrow (-k)^2 - 4(k - 1).1 = 0 \Rightarrow k^2 - 4k + 4 = 0$$

$$\Rightarrow (k - 2)^2 = 0 \Rightarrow k - 2 = 0 \Rightarrow k = 2.$$

(v) Given the roots are numerically equal but opposite in sign, the roots may be taken as $\alpha, -\alpha$ so that the sum of roots is zero

$$\Rightarrow -\frac{k}{k - 1} = 0 \Rightarrow \frac{k}{k - 1} = 0 \Rightarrow k = 0.$$

Example 4. If one of the roots of $3x^2 - 7px + 10p = 0$ be 5, find p and also the other root.

Solution. Let the roots be $\alpha, 5$.

$$\text{Then } \alpha + 5 = -\frac{-7p}{3} = \frac{7p}{3} \text{ and } \alpha \times 5 = \frac{10p}{3} \text{ i.e. } \alpha = \frac{2p}{3}$$

$$\Rightarrow \frac{2p}{3} + 5 = \frac{7p}{3} \Rightarrow 2p + 15 = 7p$$

$$\Rightarrow -5p = -15 \Rightarrow p = 3.$$

$$\therefore \alpha = \frac{2 \times 3}{3} = 2.$$

Hence, $p = 3$, and the other root is 2.

Example 5. The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. Prove that the product of the roots is $-\frac{1}{2}(a^2 + b^2)$.

Solution. The given equation is $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$

$$\Rightarrow \frac{x+b+x+a}{(x+a)(x+b)} = \frac{1}{c}$$

$$\Rightarrow (x+a)(x+b) = c(2x+a+b)$$

$$\Rightarrow x^2 + (a+b)x + ab - 2cx - ca - bc = 0$$

$$\Rightarrow x^2 + (a+b-2c)x + (ab-bc-ca) = 0.$$

$$\text{Sum of roots} = 0 \text{ (given)} \Rightarrow -\frac{a+b-2c}{1} = 0$$

$$\Rightarrow a+b-2c=0 \Rightarrow c = \frac{a+b}{2} \quad \dots(i)$$

$$\text{Product of roots} = \frac{ab-bc-ca}{1} = ab - (a+b)c$$

$$= ab - (a+b) \cdot \frac{a+b}{2} \quad \text{(using (i))}$$

$$= \frac{2ab - (a+b)^2}{2} = -\frac{a^2 + b^2}{2}.$$

Example 6. If α, β be the roots of the equation $ax^2 + bx + c = 0$, evaluate

$$(i) \alpha^3 \beta + \alpha \beta^3 \quad (ii) \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} \quad (iii) (a\alpha + b)^{-2} + (a\beta + b)^{-2}.$$

Solution. Since α, β are roots of $ax^2 + bx + c = 0$,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

$$(i) \alpha^3 \beta + \alpha \beta^3 = \alpha\beta(\alpha^2 + \beta^2) = \alpha\beta[(\alpha + \beta)^2 - 2\alpha\beta]$$

$$= \frac{c}{a} \cdot \left[\left(-\frac{b}{a} \right)^2 - 2 \cdot \frac{c}{a} \right] = \frac{c}{a} \cdot \frac{b^2 - 2ac}{a^2} = \frac{c(b^2 - 2ac)}{a^3}.$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} &= \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{\alpha\beta} = \frac{\left(-\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(-\frac{b}{a}\right)}{\frac{c}{a}} \\
 &= \frac{-b^3 + 3abc}{a^3} \cdot \frac{a}{c} = \frac{3abc - b^3}{a^2c}.
 \end{aligned}$$

(iii) Since α is root of $ax^2 + bx + c = 0$, we have $a\alpha^2 + b\alpha + c = 0$
 $\Rightarrow \alpha(a\alpha + b) = -c \Rightarrow a\alpha + b = -\frac{c}{\alpha}$.

Similarly, $a\beta + b = -\frac{c}{\beta}$.

$$\begin{aligned}
 \therefore (a\alpha + b)^{-2} + (a\beta + b)^{-2} &= \left(-\frac{c}{\alpha}\right)^{-2} + \left(-\frac{c}{\beta}\right)^{-2} = \left(\frac{\alpha}{c}\right)^2 + \left(\frac{\beta}{c}\right)^2 = \frac{\alpha^2 + \beta^2}{c^2} \\
 &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} = \frac{\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{c^2} = \frac{b^2 - 2ac}{a^2c^2}.
 \end{aligned}$$

Example 7. (i) Form an equation whose roots are $2, -\frac{1}{2}$.

(ii) Form an equation with rational coefficients, one of whose roots is $\frac{1}{3 + 2\sqrt{2}}$.

(iii) Form an equation with real coefficients one of whose roots is $-2 + i$.

Solution. (i) Since roots are $2, -\frac{1}{2}$; sum of roots $S = 2 + \left(-\frac{1}{2}\right) = \frac{3}{2}$

and product of roots $P = 2 \cdot \left(-\frac{1}{2}\right) = -1$.

The required equation is $x^2 - Sx + P = 0$ i.e. $x^2 - \frac{3}{2}x - 1 = 0$

i.e. $2x^2 - 3x - 2 = 0$.

(ii) One of the roots is $\frac{1}{3 + 2\sqrt{2}} = \frac{1}{3 + 2\sqrt{2}} \cdot \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}} = \frac{3 - 2\sqrt{2}}{9 - 8} = 3 - 2\sqrt{2}$.

Since the equation has rational coefficients, the other root is $3 + 2\sqrt{2}$.

$\therefore$ Sum of roots $S = (3 - 2\sqrt{2}) + (3 + 2\sqrt{2}) = 6$.

Product of roots $P = (3 - 2\sqrt{2})(3 + 2\sqrt{2}) = 9 - 8 = 1$.

The required equation is $x^2 - Sx + P = 0$ i.e. $x^2 - 6x + 1 = 0$.

(iii) Since one of the roots is $-2 + i$, and the equation has real coefficients, the other root must be $-2 - i$.

$\therefore$ Sum of roots $S = (-2 + i) + (-2 - i) = -4$.

Product of roots $P = (-2 + i)(-2 - i) = 4 - i^2 = 4 + 1 = 5$.

$\therefore$ Required equation is $x^2 - Sx + P = 0$ i.e. $x^2 + 4x + 5 = 0$.

Example 8. If α, β are roots of the quadratic equation $ax^2 + bx + c = 0$, form an equation whose roots are

- (i) $-\alpha, -\beta$ (ii) $\frac{1}{\alpha}, \frac{1}{\beta}$ (iii) α^2, β^2 (iv) α^3, β^3 .

Solution. Since α, β are roots of $ax^2 + bx + c = 0$,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

(i) Here, $S = (-\alpha) + (-\beta) = -(\alpha + \beta) = -\left(-\frac{b}{a}\right) = \frac{b}{a}$ and

$$P = (-\alpha)(-\beta) = \alpha\beta = \frac{c}{a}.$$

The required equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - \frac{b}{a}x + \frac{c}{a} = 0 \text{ i.e. } ax^2 - bx + c = 0.$$

(ii) Here, $S = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = -\frac{b}{c}$ and

$$P = \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{c/a} = \frac{a}{c}.$$

The required equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - \left(-\frac{b}{c}\right)x + \frac{a}{c} = 0 \text{ i.e. } cx^2 + bx + a = 0.$$

(iii) Here, $S = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{b}{a}\right)^2 - 2 \cdot \frac{c}{a} = \frac{b^2 - 2ac}{a^2}$ and

$$P = \alpha^2 \cdot \beta^2 = (\alpha\beta)^2 = \left(\frac{c}{a}\right)^2 = \frac{c^2}{a^2}.$$

The required equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - \frac{b^2 - 2ac}{a^2}x + \frac{c^2}{a^2} = 0$$

$$\text{i.e. } a^2x^2 - (b^2 - 2ac)x + c^2 = 0.$$

(iv) Here, $S = \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$$= \left(-\frac{b}{a}\right)^3 - 3 \cdot \frac{c}{a} \cdot \left(-\frac{b}{a}\right) = -\frac{b^3}{a^3} + \frac{3bc}{a^2} = \frac{3abc - b^3}{a^3} \text{ and}$$

$$P = \alpha^3 \cdot \beta^3 = (\alpha\beta)^3 = \left(\frac{c}{a}\right)^3 = \frac{c^3}{a^3}.$$

The required equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - \frac{3abc - b^3}{a^3}x + \frac{c^3}{a^3} = 0$$

$$\text{i.e. } a^3x^2 - (3abc - b^3)x + c^3 = 0.$$

Example 9. Find the equation whose roots are n times the roots of the equation $ax^2 + bx + c = 0$.

Solution. Let α, β be roots of the equation $ax^2 + bx + c = 0$, then

$$\alpha + \beta = -\frac{b}{a} \quad \dots(i) \quad \text{and} \quad \alpha\beta = \frac{c}{a} \quad \dots(ii)$$

Since the roots of the required equation are n times the roots of the given equation, roots of the required equation are $n\alpha, n\beta$.

Sum of the roots $S = n\alpha + n\beta = n(\alpha + \beta)$

$$= n \left(-\frac{b}{a} \right) \quad (\text{using (i)})$$

and product of the roots $P = n\alpha \cdot n\beta = n^2 \cdot \alpha\beta = n^2 \cdot \frac{c}{a}$ (using (ii))

$\therefore$ The required equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - \left(-\frac{nb}{a} \right)x + n^2 \frac{c}{a} = 0 \text{ or } ax^2 + nbx + n^2c = 0.$$

Example 10. Find the value of p such that the difference of the roots of the equation $x^2 - px + 10 = 0$ is

3. If the roots of $x^2 - px + 10 = 0$ are α and β , find the quadratic equation whose roots are $(\alpha + \beta)^2$ and $\alpha^2\beta - \alpha\beta^2$. (ISC 2020)

Solution. As α and β are roots of $x^2 - px + 10 = 0$, $\alpha + \beta = p$ and $\alpha\beta = 10$.

Given difference of roots is 3 i.e. $\alpha - \beta = 3$

$$\Rightarrow (\alpha - \beta)^2 = 3^2 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 9$$

$$\Rightarrow p^2 - 4 \times 10 = 9 \Rightarrow p^2 = 49 \Rightarrow p = 7, -7.$$

Roots of the equation to be formed are $(\alpha + \beta)^2$ and $\alpha^2\beta - \alpha\beta^2$.

$$(\alpha + \beta)^2 = p^2 = 49 \text{ and } \alpha^2\beta - \alpha\beta^2 = \alpha\beta(\alpha - \beta) = 10 \times 3 = 30.$$

$$\therefore S (\text{sum of roots}) = 49 + 30 = 79 \text{ and}$$

$$P (\text{product of roots}) = 49 \times 30 = 1470.$$

The required quadratic equation is $x^2 - Sx + P = 0$

$$\text{i.e. } x^2 - 79x + 1470 = 0.$$

Example 11. Let p, q be roots of $3x^2 + 6x + 2 = 0$. Form an equation whose roots are $\frac{-p^2}{q}, \frac{-q^2}{p}$.

Solution. Since p, q are roots of $3x^2 + 6x + 2 = 0$, $p + q = -\frac{6}{3} = -2$, $pq = \frac{2}{3}$.

$$\begin{aligned} \text{Here, } S &= -\frac{p^2}{q} - \frac{q^2}{p} = -\frac{p^3 + q^3}{pq} = -\frac{(p+q)^3 - 3pq(p+q)}{pq} = -\frac{(-2)^3 - 3\left(\frac{2}{3}\right)(-2)}{\frac{2}{3}} \\ &= -(-8 + 4) \cdot \frac{3}{2} = 6 \text{ and} \end{aligned}$$

$$P = \left(-\frac{p^2}{q} \right) \left(-\frac{q^2}{p} \right) = pq = \frac{2}{3}.$$

$\therefore$ Required equation is $x^2 - Sx + P = 0$ i.e. $x^2 - 6x + \frac{2}{3} = 0$

$$\text{i.e. } 3x^2 - 18x + 2 = 0.$$

Example 12. (i) If the roots of the equation $x^2 - lx + m = 0$ differ by 1, then prove that $l^2 = 4m + 1$.

(ii) Find the condition that roots of the equation $ax^2 + bx + c = 0$ may be in the ratio $p : q$.

Solution. (i) Let the roots be α and $\alpha + 1$.

$$\text{Then } \alpha + (\alpha + 1) = -\frac{(-l)}{1} = l, \quad \alpha(\alpha + 1) = \frac{m}{1} = m.$$

From first equation, we get $\alpha = \frac{l-1}{2}$. Putting this value in second,

$$\frac{l-1}{2} \left(\frac{l-1}{2} + 1 \right) = m \Rightarrow \frac{l-1}{2} \cdot \frac{l+1}{2} = m \Rightarrow l^2 - 1 = 4m$$

$$\Rightarrow l^2 = 4m + 1.$$

(ii) Since the roots of the equation $ax^2 + bx + c = 0$ are in ratio $p : q$, let the roots be $p\alpha, q\alpha$.

$$\text{Then } p\alpha + q\alpha = -\frac{b}{a}, \quad p\alpha \cdot q\alpha = \frac{c}{a}.$$

From first equation, we get $\alpha = \frac{-b}{a(p+q)}$. Putting this value in second,

$$pq \left(\frac{-b}{a(p+q)} \right)^2 = \frac{c}{a} \Rightarrow pqb^2 = ca(p+q)^2, \text{ which is the required condition.}$$

Example 13. If α, β are the roots of $x^2 + ax + b = 0$, then prove that $\frac{\alpha}{\beta}$ is a root of the equation $bx^2 + (2b - a^2)x + b = 0$.

Solution. As α, β are roots of the equation $x^2 + ax + b = 0$,

$$\alpha + \beta = -a \text{ and } \alpha\beta = b \quad \dots(i)$$

Now $\frac{\alpha}{\beta}$ is a root of the equation $bx^2 + (2b - a^2)x + b = 0$

$$\text{if } b \left(\frac{\alpha}{\beta} \right)^2 + (2b - a^2) \cdot \frac{\alpha}{\beta} + b = 0 \text{ is true}$$

$$\text{i.e. if } b(\alpha^2 + \beta^2) + (2b - a^2) \alpha\beta = 0 \text{ is true}$$

$$\text{i.e. if } b((\alpha + \beta)^2 - 2\alpha\beta) + (2b - a^2)\alpha\beta = 0 \text{ is true}$$

$$\text{i.e. if } b((-a)^2 - 2b) + (2b - a^2)b = 0 \text{ is true} \quad (\text{using (i)})$$

$$\text{i.e. if } b(a^2 - 2b) + (2b - a^2)b = 0 \text{ is true}$$

$$\text{i.e. if } 0 = 0 \text{ is true, which is true.}$$

Example 14. If the roots of the equation $ax^2 + bx + b = 0$, $b^2 \geq 4ab$, are in ratio $p : q$, prove that

$$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \sqrt{\frac{b}{a}}.$$

Solution. Since $b^2 \geq 4ab$, the roots are real. Let the roots be α, β .

$$\text{Then } \alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{b}{a}. \text{ Also } \frac{\alpha}{\beta} = \frac{p}{q} \text{ (given).}$$

$$\text{Now } \left(\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} \right)^2 = \frac{p}{q} + \frac{q}{p} + 2 = \frac{\alpha}{\beta} + \frac{\beta}{\alpha} + 2 = \frac{\alpha^2 + \beta^2 + 2\alpha\beta}{\alpha\beta} = \frac{(\alpha + \beta)^2}{\alpha\beta} = \frac{(-b/a)^2}{b/a} = \frac{b}{a}.$$

$$\text{Taking square roots of both sides, we get } \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} = \sqrt{\frac{b}{a}}. \quad (\because \text{L.H.S.} \geq 0)$$

Example 15. The roots of the equation $px^2 - 2(p+2)x + 3p = 0$ differ by 2. Find p and also the roots.

Solution. Let the roots be $\alpha, \alpha + 2$.

$$\text{Then } \alpha + \alpha + 2 = \frac{2(p+2)}{p}, \quad \alpha(\alpha + 2) = \frac{3p}{p} = 3.$$

$$\text{Now } \alpha + \alpha + 2 = \frac{2p+4}{p} = 2 + \frac{4}{p} \Rightarrow 2\alpha = \frac{4}{p} \Rightarrow \alpha = \frac{2}{p}.$$

$$\therefore \alpha(\alpha + 2) = 3 \Rightarrow \frac{2}{p} \left(\frac{2}{p} + 2 \right) = 3 \Rightarrow 3p^2 - 4p - 4 = 0$$

$$\Rightarrow (3p + 2)(p - 2) = 0 \Rightarrow p = 2, -\frac{2}{3}.$$

When $p = 2$, then $\alpha = \frac{2}{p} = 1$; so roots are 1, 3.

When $p = -\frac{2}{3}$, then $\alpha = \frac{2}{p} = -3$; so roots are -3, -1.

Example 16. Solve the equation $x^2 + px + 45 = 0$, given that the square of the difference of its roots is 144.

Solution. Let α, β be the roots of the equation $x^2 + px + 45 = 0$, then

$$\alpha + \beta = -p \text{ and } \alpha\beta = 45.$$

It is given that $(\alpha - \beta)^2 = 144$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 144 \Rightarrow (-p)^2 - 4 \times 45 = 144$$

$$\Rightarrow p^2 = 144 + 180 = 324 \Rightarrow p = 18, -18.$$

When $p = 18$, the given equation becomes $x^2 + 18x + 45 = 0$

$$\Rightarrow (x + 3)(x + 15) = 0 \Rightarrow x = -3, -15.$$

When $p = -18$, the given equation becomes

$$x^2 - 18x + 45 = 0 \Rightarrow (x - 3)(x - 15) = 0 \Rightarrow x = 3, 15.$$

Hence, the roots of the given equation are -3, -15 or 3, 15.

Example 17. If the difference of the roots of the equation $x^2 + px + q = 0$ be the same as that of the roots of $x^2 + qx + p = 0$, then prove that $p + q + 4 = 0$; $p \neq q$.

Solution. Let α, β be the roots of $x^2 + px + q = 0$ and

γ, δ be the roots of $x^2 + qx + p = 0$, then

$$\alpha + \beta = -p, \alpha\beta = q \quad \dots(i)$$

$$\text{and } \gamma + \delta = -q, \gamma\delta = p \quad \dots(ii)$$

$$\text{Given } \alpha - \beta = \gamma - \delta \Rightarrow (\alpha - \beta)^2 = (\gamma - \delta)^2$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$$

$$\Rightarrow (-p)^2 - 4q = (-q)^2 - 4p \quad \text{(using (i) and (ii))}$$

$$\Rightarrow p^2 - 4q = q^2 - 4p$$

$$\Rightarrow p^2 - q^2 + 4p - 4q = 0$$

$$\Rightarrow (p - q)(p + q) + 4(p - q) = 0$$

$$\Rightarrow (p - q)(p + q + 4) = 0 \text{ but } p \neq q \text{ i.e. } p - q \neq 0$$

$$\Rightarrow p + q + 4 = 0.$$

Example 18. (i) If the ratio of the roots of the equation $x^2 + px + q = 0$ is equal to the ratio of the roots of the equation $x^2 + bx + c = 0$, then prove that $p^2c = b^2q$.

(ii) If α, β are roots of the equation $x^2 + px + q = 0$ and γ, δ are roots of the equation $x^2 + bx + c = 0$ such that $\alpha, \beta, \gamma, \delta$ are in G.P., then prove that $p^2c = b^2q$.

Solution. (i) Let α, β be roots of the equation $x^2 + px + q = 0$, then

$$\alpha + \beta = -p \text{ and } \alpha\beta = q \quad \dots(1)$$

Let γ, δ be roots of the equation $x^2 + bx + c = 0$, then

$$\gamma + \delta = -b \text{ and } \gamma\delta = c \quad \dots(2)$$

According to given, $\frac{\alpha}{\beta} = \frac{\gamma}{\delta}$

$$\Rightarrow \frac{\alpha - \beta}{\alpha + \beta} = \frac{\gamma - \delta}{\gamma + \delta} \quad (\text{using componendo and dividendo})$$

$$\Rightarrow \frac{(\alpha - \beta)^2}{(\alpha + \beta)^2} = \frac{(\gamma - \delta)^2}{(\gamma + \delta)^2} \Rightarrow \frac{(\alpha + \beta)^2 - 4\alpha\beta}{(\alpha + \beta)^2} = \frac{(\gamma + \delta)^2 - 4\gamma\delta}{(\gamma + \delta)^2}$$

$$\Rightarrow \frac{(-p)^2 - 4q}{(-p)^2} = \frac{(-b)^2 - 4c}{(-b)^2} \quad (\text{using (1) and (2)})$$

$$\Rightarrow \frac{p^2 - 4q}{p^2} = \frac{b^2 - 4c}{b^2} \Rightarrow 1 - \frac{4q}{p^2} = 1 - \frac{4c}{b^2}$$

$$\Rightarrow -\frac{4q}{p^2} = -\frac{4c}{b^2} \Rightarrow \frac{q}{p^2} = \frac{c}{b^2} \Rightarrow p^2c = b^2q.$$

(ii) Given $\alpha, \beta, \gamma, \delta$ are in G.P. Let r be the common ratio of the G.P.,

$$\text{then } \frac{\alpha}{\beta} = \frac{1}{r} \text{ and } \frac{\gamma}{\delta} = \frac{1}{r}$$

$$\Rightarrow \frac{\alpha}{\beta} = \frac{\gamma}{\delta}, \text{ now proceed as in part (i).}$$

Example 19. The coefficient of x in the equation $x^2 + px + q = 0$ was taken as 17 in place of 13 and thus its roots were found to be -2 and -15 . Find the roots of the original equation.

Solution. By given conditions, -2 and -15 are roots of the equation $x^2 + 17x + q = 0$.

$$\text{The product of roots} = (-2)(-15) = \frac{q}{1} \Rightarrow q = 30.$$

$$\text{Therefore, the original equation is } x^2 + 13x + 30 = 0$$

$$\Rightarrow (x + 10)(x + 3) = 0 \Rightarrow x = -3, -10.$$

Hence, the roots of the original equation are $-3, -10$.

Example 20. If one root of the equation $ax^2 + bx + c = 0$ is the square of the other, prove that

$$b^3 + ac(c + a) = 3abc.$$

Solution. Let the roots of the given equation $ax^2 + bx + c = 0$ be α and α^2 .

$$\text{Then } \alpha + \alpha^2 = -\frac{b}{a} \quad \dots(i) \quad \text{and} \quad \alpha \cdot \alpha^2 = \frac{c}{a} \Rightarrow \alpha^3 = \frac{c}{a} \quad \dots(ii)$$

Cubing both sides of (i), we get

$$\alpha^3 + \alpha^6 + 3\alpha \cdot \alpha^2 (\alpha + \alpha^2) = -\frac{b^3}{a^3}$$

$$\Rightarrow \alpha^3 + (\alpha^3)^2 + 3\alpha^3 (\alpha + \alpha^2) = -\frac{b^3}{a^3}$$

$$\Rightarrow \frac{c}{a} + \frac{c^2}{a^2} + 3 \cdot \frac{c}{a} \left(-\frac{b}{a} \right) = -\frac{b^3}{a^3} \quad (\text{using (i) and (ii)})$$

$$\Rightarrow a^2c + ac^2 - 3abc = -b^3 \quad (\text{multiplying by } a^3)$$

$$\Rightarrow b^3 + ac(c + a) = 3abc, \text{ as required.}$$

Example 21. If one root of the quadratic equation $ax^2 + bx + c = 0$ is equal to the n th power of the other root, then show that $(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0$.

Solution. Let the roots of the given equation $ax^2 + bx + c = 0$ be α and α^n .

$$\text{Then } \alpha \cdot \alpha^n = \frac{c}{a} \Rightarrow \alpha^{n+1} = \frac{c}{a} \Rightarrow \alpha = \left(\frac{c}{a}\right)^{\frac{1}{n+1}} \quad \dots(i)$$

Since α is a root of the given equation $ax^2 + bx + c = 0$,

$$\begin{aligned} a\alpha^2 + b\alpha + c &= 0 \Rightarrow a\alpha + b + \frac{c}{\alpha} = 0 \\ \Rightarrow a \left(\frac{c}{a}\right)^{\frac{1}{n+1}} + b + c \cdot \left(\frac{a}{c}\right)^{\frac{1}{n+1}} &= 0 \quad \text{(using (i))} \\ \Rightarrow \left(a^{n+1} \cdot \frac{c}{a}\right)^{\frac{1}{n+1}} + b + \left(c^{n+1} \cdot \frac{a}{c}\right)^{\frac{1}{n+1}} &= 0 \\ \Rightarrow (a^n c)^{\frac{1}{n+1}} + b + (c^n a)^{\frac{1}{n+1}} &= 0. \end{aligned}$$

Example 22. If the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root and $a \neq b$, then prove that $a + b + 1 = 0$.

Solution. Let α be the common root of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$, then

$$\alpha^2 + a\alpha + b = 0 \quad \dots(i)$$

$$\text{and } \alpha^2 + b\alpha + a = 0 \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$(a - b)\alpha + (b - a) = 0$$

$$\Rightarrow (a - b)(\alpha - 1) = 0 \Rightarrow \alpha = 1 \quad (\because a \neq b)$$

Substituting $\alpha = 1$ in (i), we get

$$1^2 + a \times 1 + b = 0 \Rightarrow a + b + 1 = 0$$

Example 23. If the equations $3x^2 + px + 1 = 0$ and $2x^2 + qx + 1 = 0$ have a common root, show that $2p^2 + 3q^2 - 5pq + 1 = 0$.

Solution. Let α be the common root of $3x^2 + px + 1 = 0$ and $2x^2 + qx + 1 = 0$, then

$$3\alpha^2 + p\alpha + 1 = 0 \quad \dots(i)$$

$$\text{and } 2\alpha^2 + q\alpha + 1 = 0 \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$\alpha^2 + (p - q)\alpha = 0 \Rightarrow \alpha + p - q = 0 \quad (\because \alpha \neq 0)$$

$$\Rightarrow \alpha = q - p.$$

Substituting this value of α in (i), we get

$$3(q - p)^2 + p(q - p) + 1 = 0$$

$$\Rightarrow 3(q^2 + p^2 - 2pq) + pq - p^2 + 1 = 0$$

$$\Rightarrow 2p^2 + 3q^2 - 5pq + 1 = 0.$$